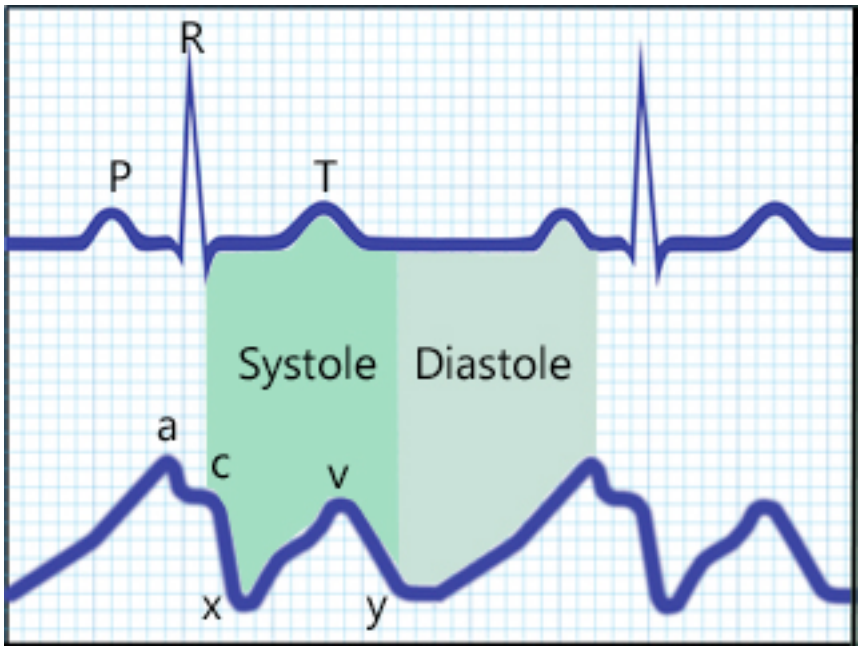
CVP Waveform *Prodigy Anesthesia*

The central venous pressure (CVP) waveform is primarily used to assess a patient's fluid volume status, but it can also be used to help monitor dysrhythmias, right-sided heart valve defects, tamponade, and ischemia. The normal range for the CVP is 2 mm Hg to 7 mm Hg.

The normal CVP waveform consists of three positive deflections (the 'a', 'c', and 'v' waves) and two negative deflections (the 'x' descent and 'y' descent).

The 'a' wave corresponds with the increase in right atrial pressure as the atrium contracts. The 'c' wave is produced as the closed tricuspid valve bulges into the right atrium at the onset of ventricular systole. The 'x' descent is produced during ventricular systole as the atrium relaxes and right atrial pressure decreases rapidly. The 'v' wave occurs during filling of the right atrium. The 'y' descent represents the drop in atrial pressure when the tricuspid valve opens and the atrium empties.



If a patient is hypovolemic, a significant increase in intravascular volume will demonstrate only a small change in the central venous pressure. If the patient is fluid overloaded, a small increase in intravascular volume can increase the central venous pressure significantly.

Patients in atrial fibrillation will not demonstrate an 'a' wave, but will have a prominent 'c' wave on the CVP waveform. Patients with tricuspid regurgitation will demonstrate a prominent 'c-v' wave as the regurgitant volume of blood crosses the tricuspid valve in a retrograde fashion.

Cardiac tamponade is associated with an overall elevation of the CVP waveform and absence of the 'y' descent.

Atrioventricular dissociation may produce cannon 'a' waves when the atrium contracts against a closed tricuspid valve.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 324.

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